

SMid Cap Industrials

Gemba Series Key Takeaways: Physical AI Inflects Automation Demand as Pilots Move to Plans

We hosted a Gemba Series webinar with Andrea Bonfanti, VP Technology and Global Head of Technology and Innovation for Robotics AI and Data Center Industries at ABB Robotics (Not Covered). Our discussion focused on the rapidly evolving robotics and humanoid landscape, including industrial automation demand, Physical AI, deployment challenges, and where value creation may emerge over the coming years. Andrea's view is that physical AI has inflected automation demand toward adaptability including potentially humanoids, yet scaled adoption hinges on resolving actuator and supply chain bottlenecks, reducing hardware costs, and forging partnerships that unlock proprietary shop-floor data.

- **Adaptability Displaces Ten-Year Line Economics:** Andrea believes automation demand has inflected over the past six months, with pilots proliferating across structured and increasingly semi-structured environments where physical AI now carries the value proposition. Geopolitical friction, labor costs, and reshoring were cited as the drivers justifying capex, while compressed product life cycles mean customers no longer specify lines for a decade of use. Andrea believes this adaptability requirement is the central shift, extending beyond cobots and conventional industrial robotics to humanoids. Deployment remains concentrated in logistics but is broadening across SKUs, with automotive material kitting and flexible material handling following and data center server rack inspection emerging as a newer lever.
- **Brownfield Favors Humanoids: Purpose-Built Wins on Speed and Greenfield Complexity:** Industrial customers buy KPIs, with throughput, speed, and reliability governing the decision. On that basis, Andrea believes purpose-built robots remain the choice for high-speed execution, greenfield installations, and complex specialized processes. The humanoid case rests instead on the estimate that the majority of shop floors are built for humans, leaving narrow aisles that a cobot mounted on an AMR cannot access. Brownfield facilities that cannot be economically renovated therefore represent the primary opportunity, as customers will fund a >\$100k robot before retrofitting a plant, making plug-and-play penetration the realistic path.
- **Actuators Remain a Binding Constraint:** Actuators and motion control represent ~50-60% of total humanoid bill of materials (BOM), making actuation both the principal cost driver and the principal supply chain bottleneck, in Andrea's view. Design is the harder problem, requiring torque density within minimal packaging, and although progress is evident, backlogs persist across the vendor base. On software, vision-language-action (VLA) models were described as sufficient to execute certain tasks, yet reliability remains a gap given limited capacity to process multiple data modalities concurrently. Another bottleneck highlighted by Andrea is tactile sensing and bionic end effectors, where some dexterous hands are lasting only a few hundred hours.
- **No Company Runs This Race Alone:** Andrea believes the long-run winners are the OEMs playing for scale rather than first-cycle ROI, harnessing partnerships because no single participant can build and scale the full stack quickly enough to be competitive. Orchestrators that aggregate the best operating data are positioned for advantage through fleet intelligence, with

North America Small and Mid Cap Industrials

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brains increasingly hardware-agnostic yet he points toward a hybrid outcome pairing the best brain with the best connection to actuation. Critically, industrial robots cannot be trained from internet data, as decades of domain knowledge in welding, polishing, and assembly sit unstructured and isolated on the operating floor. Access is therefore a moat, and end-customer partnerships were cited as the template alongside automotive collaborations.

- **Cost Must Halve Before Scale:** Western humanoids sit at ~\$100k today against Andrea's requirement of ~\$50k or lower for ROI to clear, with hardware the primary lever and quality non-negotiable given that units cannot fail every couple thousand hours. Localized and automated actuator production, partnerships with motion specialists, and use-case-specific optimization were cited as the areas to drive cost down, noting that shoulder and elbow actuators face lighter loads than knees. EV and automotive entrants are bringing actuator volume which should help normalize pricing over time, in Andrea's view.

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